SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Technical Presentation

Code of Conduct:

I Vasantha Nithi D Y (Name / Reg No) certify that this submission is my original work and that I have adhered
to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Technical Presentation

1. Reactive Agent-Based Traffic Control System
Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis
Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations
Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System
Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management
Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

B.TECH ARTIFICIAL INTELLIGENCE AND DATA SCIENCE| CO5 AT2

Multi-Agent System *for Disaster Management*

Intelligent Agent Coordination for Real-Time Disaster Detection and Response

AI Course Technical Presentation **Vasantha Nithi D Y** **192424137**

SLIDE 2

Problem & Proposed Solution

The Problem

⚡ Speed

Disasters require **fast detection** and coordinated response — every second counts.

📡 Data Overload

Massive real-time data streams from **sensors, satellites, weather systems** and emergency reports.

🧠 Human Limits

Human-only decision-making is **too slow** for rapidly evolving disaster scenarios.

Proposed Solution

A **Multi-Agent System (MAS)** where specialized AI agents collaborate, share information, and act autonomously to detect disasters and coordinate emergency response in real time.

SLIDE 3

Multi-Agent Architecture

Sensor Agents

- Collect real-time data from IoT sensors, cameras, satellites, and weather systems
- Continuously stream environmental readings to the shared data pipeline
- Operate autonomously and in parallel across multiple geographic zones

Decision Agents

- Analyze incoming sensor data and assess disaster severity
- Combine information from multiple sources to determine appropriate actions
- Dynamically update decisions as the situation evolves

Communication Agents

- Send alerts to emergency services, authorities, and response teams
- Coordinate information flow between all stakeholders
- Distribute response instructions and situation updates in real time

Data Sources

Sensor Agents

Shared Pipeline

Decision Agents

Agents coordinate through a **shared data pipeline** and **distributed processing** — enabling parallel, scalable decision-making.

SLIDE 4

Coordination & Real-Time Processing

Continuous Monitoring

Sensor agents monitor environmental conditions — temperature, seismic activity, flood levels, and more — around the clock.

Distributed Stream Processing

Incoming data is processed in real time using distributed frameworks such as **Apache Spark / PySpark**, enabling high-throughput, low-latency analysis.

Multi-Source Data Fusion

Decision agents combine sensor streams, historical records, and satellite imagery to build a unified situational picture.

Dynamic Adaptation

Agents continuously share updates and **adapt decisions** as disaster conditions change, enabling real-time replanning.

Alert Distribution

Communication agents dispatch targeted alerts and response instructions to relevant emergency units immediately.

SLIDE 5

Benefits, Challenges & Conclusion

✅ Benefits

- → Faster disaster detection
- → Scalable processing of large real-time data
- → Improved coordination across response units
- → Real-time, data-driven decision-making
- → Significantly reduced emergency response time

⚠️ Challenges

Data Reliability

Sensor failures and noisy data can compromise situational awareness.

Communication Failures

Network disruptions may interrupt agent coordination during disasters.

Coordination Complexity

Managing consensus and conflict resolution among many agents is non-trivial.

Privacy & Security

Sensitive data from surveillance and sensors must be protected.

> **Conclusion:** A Multi-Agent System provides a **scalable and intelligent** approach for disaster management — enabling specialized AI agents to collaborate, process real-time data, and support faster emergency response.

GEO TAG PHOTO: